



SURYA THE GLOBAL SCHOOL

CONCEPT REINFORCEMENT MATERIAL



SUBJECT: MATHEMATICS

GRADE: VII

DAY-1(29-9-25)

Number System:

Integers : All positive and negative whole numbers, & zero

Ex : -3,-2,-1,0,1,2,3,4,.....

Whole Numbers : All positive integers, & zero

Ex : 0,1,2,3,4,.....

Natural Numbers : All positive integers, excluding zero

Ex : 1,2,3,4,.....

Operations on Integers:

Adding Integers of the same sign

$$\oplus + \oplus = \oplus$$

$$\ominus + \ominus = \ominus$$

Add the absolute value of each number and the result gets the same sign as the addends

$$10 + 8 = 18$$

$$-6 + -9 = -15$$

Adding Integers of opposite signs

$$\ominus + \oplus = \oplus$$

$$\oplus + \ominus = \ominus$$

Subtract the smaller absolute value from the larger and the result gets the sign from the larger

$$-7 + 13 = 6$$

$$-16 + 5 = -11$$

Subtracting Integers

$$13 - -5 =$$

$$-9 - -3 =$$

Convert the subtracted number to its opposite sign and add the numbers

$$13 + 5 = 18$$

$$-9 + 3 = -6$$

Multiplying and Dividing Integers

$$\oplus \times \oplus = \oplus$$

$$\oplus \times \ominus = \ominus$$

$$\ominus \times \ominus = \oplus$$

$$\ominus \times \oplus = \ominus$$

If the signs of the integers are the **same**, the product/quotient is **positive**

If the signs of the integers are **different**, the product/quotient is **negative**

$$-6 \times -4 = 24$$

$$-5 \times 7 = -35$$

$$-18 \div -3 = 6$$

$$12 \div -4 = -3$$

DAY-2(30-9-25)

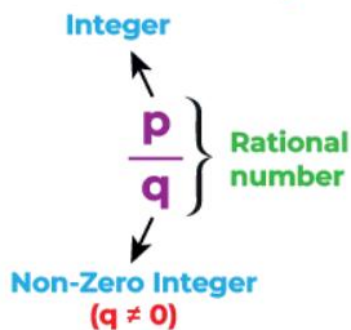
Properties of Integers:

Property	Definition	Example
Closure	Sum, difference, or product of any 2 integers is an integer (not always for division)	$-4 + 6 = 2$; $3 \times -2 = -6$
Commutative	Order doesn't change answer (addition, multiplication)	$5 + 8 = 8 + 5$; $-3 \times 7 = 7 \times -3$
Associative	Grouping doesn't change answer (addition, multiplication)	$(1 + 2) + 3 = 1 + (2 + 3)$
Distributive	Multiply over addition/subtraction	$2 \times (3 + 4) = 2 \times 3 + 2 \times 4$
Identity	Special number keeps integer same: 0 for addition, 1 for multiplication	$a + 0 = a$; $a \times 1 = a$

DAY-3(5-10-25)

Rational numbers:

Rational numbers can be represented as:



Types of Rational numbers:

- **Positive Rational Numbers** → both numerator & denominator same sign (e.g., $\frac{5}{7}$, $\frac{-9}{-11}$)
- **Negative Rational Numbers** → numerator & denominator have opposite signs (e.g., $\frac{-3}{8}$, $\frac{7}{-12}$)

Operations on Rational Numbers:

Addition:

$$\frac{2}{3} + \frac{5}{4}$$

Step 1: Since the denominator are not same,
we need to take LCM.

$$\text{LCM (4, 3)} = 12$$

Step 2: Convert to the same denominators.

$$\frac{2}{3} + \frac{5}{4} = \left(\frac{2}{3}\right) \times \left(\frac{4}{4}\right) + \left(\frac{5}{4}\right) \times \left(\frac{4}{4}\right)$$

Add the numerators.

$$= \frac{8}{12} + \frac{15}{12}$$

Keep the denominator same.

$$= \frac{23}{12}$$

Subtraction:

Subtract the numerators.

$$\frac{7}{3} - \frac{5}{3} = \frac{7-5}{3} = \frac{2}{3}$$

Keep the denominator same.

$$\frac{2}{3} - \frac{1}{4}$$

Step 1: Since the denominator are not same,
we need to take LCM.

$$\text{LCM (3, 4)} = 12$$

Step 2: Convert to the same denominators.

$$\frac{2}{3} - \frac{1}{4} = \left(\frac{2}{3}\right) \times \left(\frac{4}{4}\right) - \left(\frac{1}{4}\right) \times \left(\frac{3}{3}\right)$$

Subtract the numerators.

$$= \left(\frac{8}{12}\right) - \left(\frac{3}{12}\right)$$

Keep the denominator same.

$$= \frac{(8-3)}{12}$$

$$= \frac{5}{12}$$

DAY-5(7-10-25)

Multiplication:

Multiply numerators. **Simplify**
÷2

$$\frac{4}{6} \times \frac{8}{3} = \frac{32}{18} = \frac{16}{9}$$

Multiply denominators. **÷2**

Division:

$$\frac{2}{3} \div \frac{1}{5} = \frac{2}{3} \times \text{reciprocal of } \left(\frac{1}{5}\right)$$

Multiply numerators.

$$= \frac{2}{3} \times \frac{5}{1}$$

Multiply denominators.

$$= \frac{10}{3}$$

Simplify

$$= 3\frac{1}{3}$$

DAY-6(8-10-25)

Factors and Multiples:

- **Factor:** A number that divides another number exactly.
Example: Factors of 12 → 1, 2, 3, 4, 6, 12.
- **Multiple:** The product of a number and any whole number.
Example: Multiples of 4 → 4, 8, 12, 16...
- **Prime Number:** A number with only two factors: 1 and itself.
Example: 2, 3, 5, 7.
- **Composite Number:** A number with more than two factors.
Example: 4, 6, 8, 9.

DAY-7(9-10-25)

Decimals:

A decimal number is used to represent fractions and parts of a whole using a decimal point .

Examples: 0.5, 2.75, 13.008

- Whole number part → left of the decimal point
- Decimal (fractional) part → right of the decimal point

Whole Number Part				Decimal Part			
Thousands	Hundreds	Tens	Ones	Decimal Point	Tenths	Hundredths	Thousandths
7	5	9	4	.	1	6	3

Operations on Decimal:

Addition

$$\begin{array}{r} 29.50 \\ + 07.00 \\ \hline 36.50 \end{array}$$

Subtraction

$$\begin{array}{r} 29.50 \\ - 07.00 \\ \hline 22.50 \end{array}$$

DAY-8(10-10-25)

Decimal Divided by a Whole Number

Ex) $24.36 \div 3$

3 → divisor

24.36 → dividend

8.12

$$\begin{array}{r} 08.12 \\ 3 \overline{)24.36} \\ \underline{-24} \\ 0.3 \\ \underline{-0.3} \\ .06 \\ \underline{-0.6} \\ 0 \end{array}$$

Decimal Divided by a Decimal

Ex) $7.68 \div 0.4$

This divisor can not be a decimal
Multiply the divisor and the dividend by enough of ten
 $0.4 \overline{)7.68} \xrightarrow{\times 10} 4 \overline{)76.8}$

19.2

$$\begin{array}{r} 19.2 \\ 4 \overline{)76.8} \\ \underline{-4} \\ 36 \\ \underline{-36} \\ 08 \\ \underline{-8} \\ 0 \end{array}$$

$$\begin{array}{r} 1375 \\ \times 55 \\ \hline 75625 \end{array}$$

Forget about Decimal points
and Multiply

$$\begin{array}{r} 13.75 \leftarrow 2 \text{ d.p.} \\ \times 5.5 \leftarrow 1 \text{ d.p.} \\ \hline 75.625 \leftarrow 1 \text{ d.p.} \end{array}$$

Decimal places of the First Number
+
Decimal places of the Second Number
=
Decimal places of the Third Number

DAY-9(11-10-25)

Fractions: Definition: A number that represents a part of a whole.

Example: $\frac{3}{4}$ means 3 parts out of 4.

Types of fractions	Definition	Example
Unit fractions	Fractions with numerator 1 .	$\frac{1}{7}$
Proper Fractions	Fractions in which the numerator is less than the denominator.	$\frac{2}{7}$
Improper Fractions	Fractions in which the numerator is more than or equal to the denominator.	$\frac{5}{3}$
Mixed Fractions	Mixed fractions consist of a whole number along with a proper fraction.	$8\frac{2}{3}$
Like Fractions	Fractions with the same denominators.	$\frac{1}{4}$ and $\frac{3}{4}$
Unlike Fractions	Fractions with different denominators.	$\frac{1}{3}$ and $\frac{3}{4}$
Equivalent Fractions	Fractions that have the same value after being simplified or reduced.	$\frac{6}{4}$ and $\frac{12}{8}$

DAY-10(12-10-25)

FRACTION OPERATIONS

+

-

Adding or subtracting fractions with common denominators

$$\frac{1}{5} + \frac{2}{5} = \frac{3}{5}$$

Add (or subtract) the numerators, denominator stays the same, simplify, if possible.

+

-

Adding or subtracting fractions with different denominators

$$\frac{3}{4} - \frac{1}{3} = \frac{9}{12} - \frac{4}{12} = \frac{5}{12}$$

Find least common denominator, subtract (or add) the numerators, simplify, if possible.

×

Multiplying fractions

$$\frac{2}{3} \times \frac{3}{6} = \frac{6}{18} = \frac{1}{3}$$

Multiply the numerators, multiply the denominators, simplify, if possible.

÷

Dividing fractions

$$\frac{3}{5} \div \frac{2}{3} = \frac{3}{5} \times \frac{3}{2} = \frac{9}{10}$$

Multiply the first fraction by the reciprocal of the second fraction, simplify, if possible.

DAY-11(13-10-25)

Measurement:(conversion of units)

METRIC UNITS OF MEASURE		
Length	Capacity	Weight (mass)
1 kilometer (km) = 1000 meters (m)	1 kiloliter (kl) = 1000 liters (L)	1 kilogram (kg) = 1000 grams(g)
1 hectometer (hm) = 100 m	1 hectoliter (hl) = 100 m	1 hectogram (hg) = 100 g
1 dekameter (dam) = 10 m	1 dekaliter (dal) = 10 m	1 dekagram (dag) = 10 g
1 meter (m) = 1 m	1 liter (L) = 1 L	1 gram (g) = 1 g
1 decimeter (dm) = 0.1 m	1 deciliter (dl) = 0.1 L	1 decigram (dg) = 0.1 g
1 centimeter (cm) = 0.01 m	1 centiliter (cl) = 0.01 L	1 centigram (cg) = 0.01 g
1 millimeter (mm) = 0.001 m	1 milliliter (ml) = 0.001 L	1 milligram (mg) = 0.001 g

DAY-12(14-10-25)

Ratio :

Ratio: A ratio is a way to compare two quantities of the same kind by division.

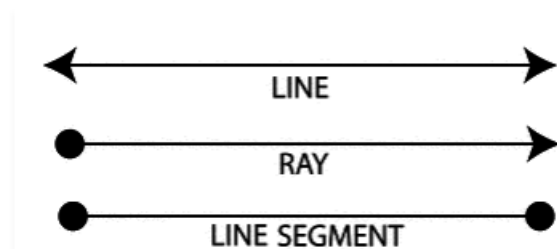
- Written as **a : b** (read as “a is to b”).
- **Example:** In a basket, there are **8 apples** and **4 oranges**.
- Ratio of apples to oranges = 8:4=2:1
- This means → there are 2 apples for every 1 orange

Geometry :

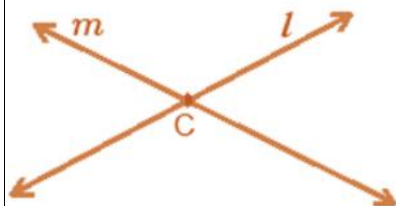
Line: If we take a point and draw a straight path that extends endlessly on both the sides, then the straight path is called as a line.

Ray A ray is a part of a line with one endpoint.

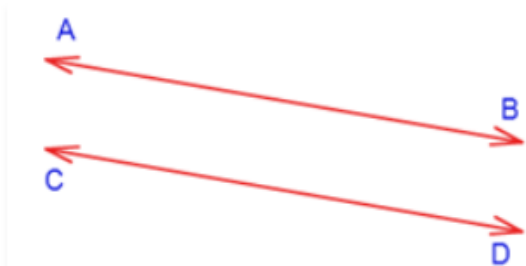
Line segment: A line segment is a part of a line with two endpoints.



Intersecting Lines: If two lines have a common point, then they are said to be intersecting.

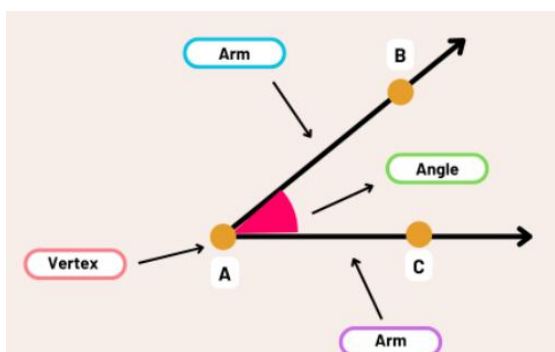


Parallel lines: If two lines have no common point, then they are said to be parallel. Here AB and CD are parallel lines.



DAY-13(15-10-25)

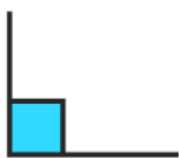
Angle:



- **Types of Angles:**



ACUTE ANGLE
Less than 90°



RIGHT ANGLE
Exact 90°



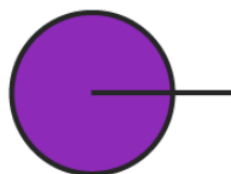
OBTUSE ANGLE
Greater than 90°
and less than 180°



STRAIGHT ANGLE
Exact 180°



REFLEX ANGLE
Greater than 180°



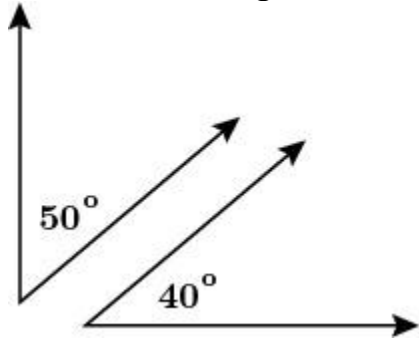
FULL ANGLE
Exact 360°

Complementary Angles:

Two angles whose sum is 90° are called complementary angles.

Example: $50^\circ + 40^\circ = 90^\circ$

$\therefore 50^\circ$ and 40° angles are complementary angles.

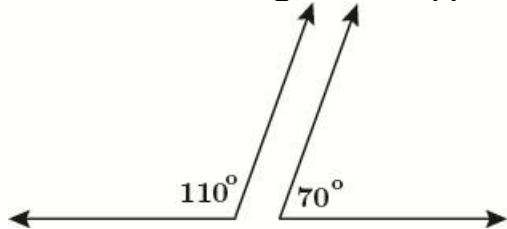


Supplementary angles

- Two angles whose sum is 180° are called supplementary angles.

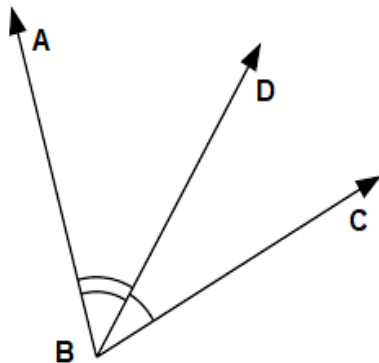
Example: $110^\circ + 70^\circ = 180^\circ$

$\therefore 110^\circ$ and 70° angles are supplementary angles.



Adjacent Angles:

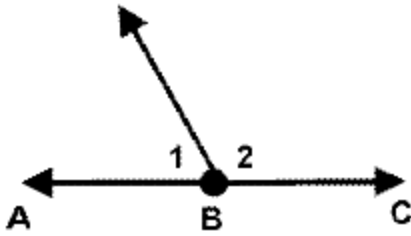
- Two angles are adjacent, if they have
 - A common vertex
 - A common arm
 - Their non-common arms on different sides of the common arm.



- Here $\angle ABD$ and $\angle DBC$ are adjacent angles.

Linear Pair:

- Linear pair of angles are adjacent angles whose sum is equal to 180° .

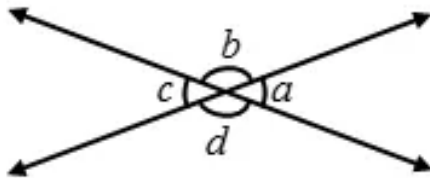


Here, 1 and 2 are linear pair of angles.

DAY-14(16-10-25)

Vertically Opposite Angles:

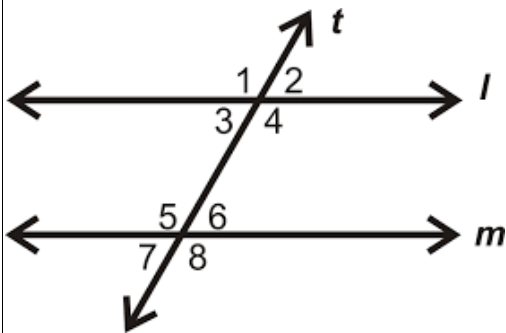
- Vertically opposite angles are formed when two straight lines intersect each other at a common point.
- Vertically opposite angles are equal.



Here, the following pairs of angles are vertically opposite angles.

- (i) a and c
- (ii) b and d

Transversal of Parallel Lines:



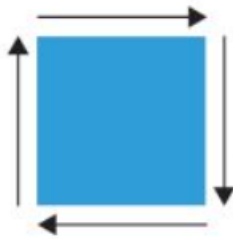
- If a transversal intersects two parallel lines, then each pair of corresponding angles is equal.
 - (i) $\angle 1 = \angle 5$ (ii) $\angle 2 = \angle 6$
 - (iii) $\angle 3 = \angle 7$ (iv) $\angle 4 = \angle 8$

- If a transversal intersects two parallel lines, then each pair of alternate interior angles is equal.
(i) $\angle 3 = \angle 6$ (ii) $\angle 4 = \angle 5$
- If a transversal intersects two parallel lines, then each pair of interior angles on the same side of the transversal is supplementary.
(i) $\angle 3 + \angle 5 = 180^\circ$ (ii) $\angle 4 + \angle 6 = 180^\circ$

DAY-15(17-10-25)

Perimeter & Area:

- **Perimeter:** Distance around a shape.



Perimeter of a rectangle = 2 x (length + breadth)

Example:

Length = 12 cm, Breadth = 8 cm

$$\text{Perimeter} = 2 \times (12 + 8) = 2 \times 20 = 40 \text{ cm}$$

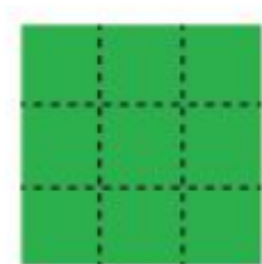
Perimeter of a square = 4 × side

Example:

Side = 7 cm

$$\text{Perimeter} = 4 \times 7 = 28 \text{ cm}$$

Area: Space inside a shape.



Area of a square = side $\times$ side.

Example:

Side = 6 cm

$$\text{Area} = 6 \times 6 = 36 \text{ cm}^2$$

Area of Rectangle = Length $\times$ Breadth

Example:

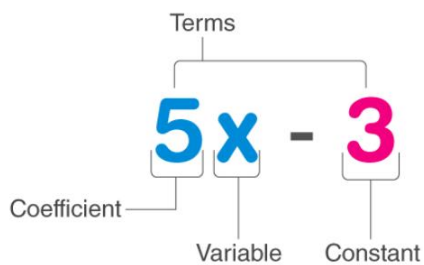
Length = 12 cm, Breadth = 5 cm

$$\text{Area} = 12 \times 5 = 60 \text{ cm}^2$$

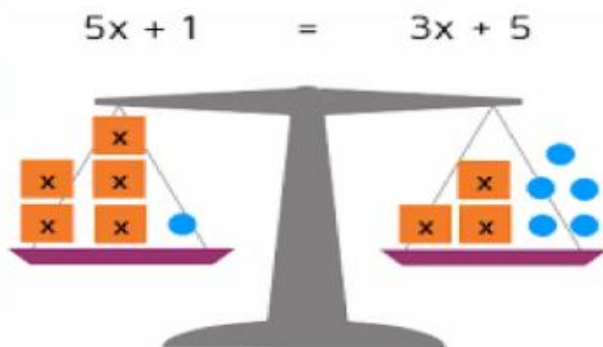
DAY-16(18-10-25)

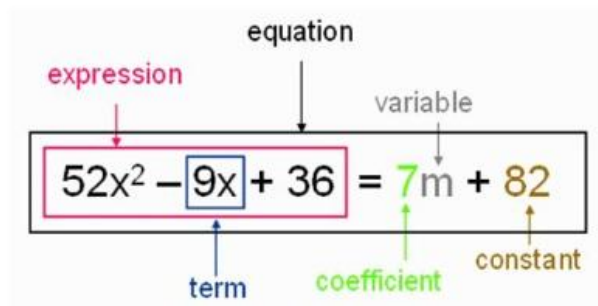
Algebra :

ALGEBRAIC EXPRESSIONS



Simple Equations: An equation has an equal sign =.

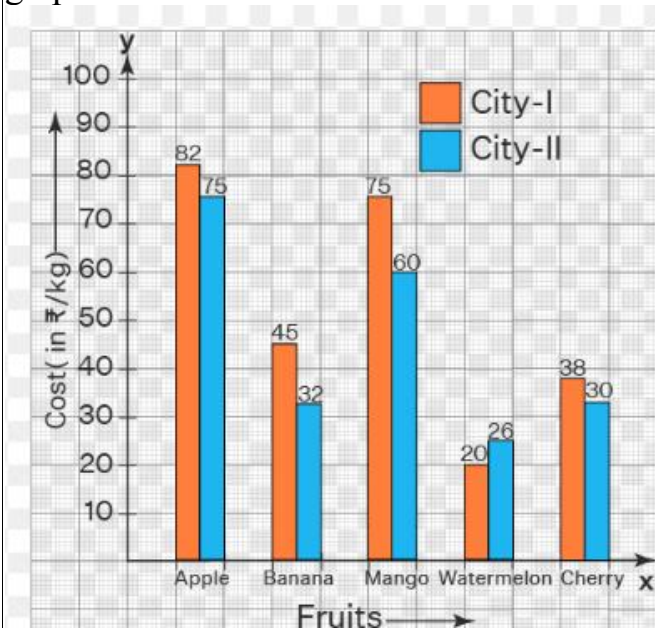




DAY-17(19-10-25)

Data Handling:

Double bar graph: A double bar graph is used to display two sets of data on the same graph.



Mean (Average): The mean is the sum of all values divided by the number of values.

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}$$

Example:

The marks obtained by 5 students in a test are: 12, 18, 20, 15, and 25.
Find the mean of their marks.

Solution:

$$\text{Mean} = \frac{\text{Sum of observations}}{\text{Number of observations}}$$

$$\text{Sum of marks} = 12 + 18 + 20 + 15 + 25 = 90$$

$$\text{Number of students} = 5$$

$$\text{Mean} = \frac{90}{5} = 18$$

Median (Middle Value): The **median** is the middle value when the data is arranged in order.

- If the number of values is **odd** → middle value.
- If the number of values is **even** → average of the two middle values.

Example:1 Find the median of the numbers: 7, 3, 9, 5, 11

Solution:

Step 1: Arrange in ascending order → 3, 5, 7, 9, 11

Step 2: Number of observations = 5 (odd)

Middle value = 3rd number = 7

Answer: Median = 7

Example:2 Find the median of the numbers:

12, 6, 10, 4

Solution:

Step 1: Arrange in ascending order → 4, 6, 10, 12

Step 2: Number of observations = 4 (even)

$$\text{Median} = \frac{\text{Middle two terms}}{2}$$

Middle terms = 6 and 10

$$\text{Median} = \frac{6 + 10}{2} = \frac{16}{2} = 8$$

Mode (Most Frequent Value): The **mode** is the value that occurs most often.

Example 1 (Single Mode):

Find the mode of:

4, 7, 9, 4, 3, 4, 5, 7

Solution:

- Number 4 occurs **3 times**
- Number 7 occurs **2 times**
- Other numbers occur once

So, **Mode = 4**
